Freddy erazo

1. Introduction

This report details a comprehensive data mining project conducted on a retail transaction dataset to uncover consumer behavior patterns and optimize business operations. The dataset consists of over 38,000 individual purchase records spanning a two-year period from 2014 to 2015. The primary objective was to move beyond raw data to actionable intelligence using the Knowledge Discovery in Databases (KDD) process.

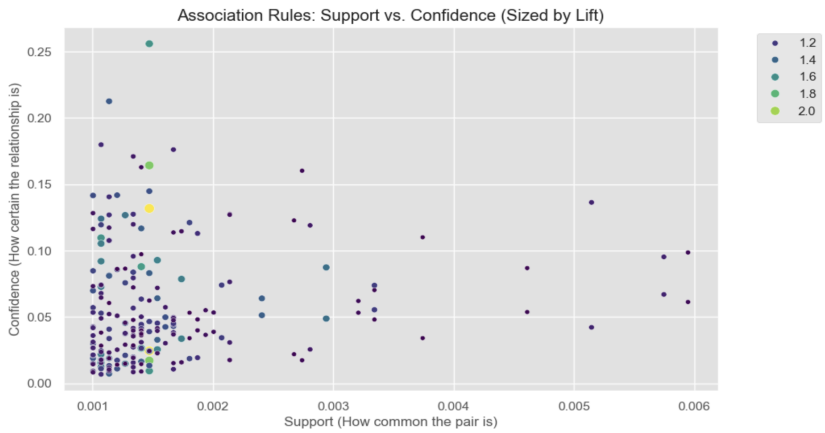
The analysis was guided by several discovery questions: What are the primary "anchor" products driving store traffic? Does the store serve as a major weekly shopping destination or a high-frequency "top-off" location? Finally, can we segment the customer base into distinct behavioral profiles to allow for targeted marketing strategies?

2. Methods

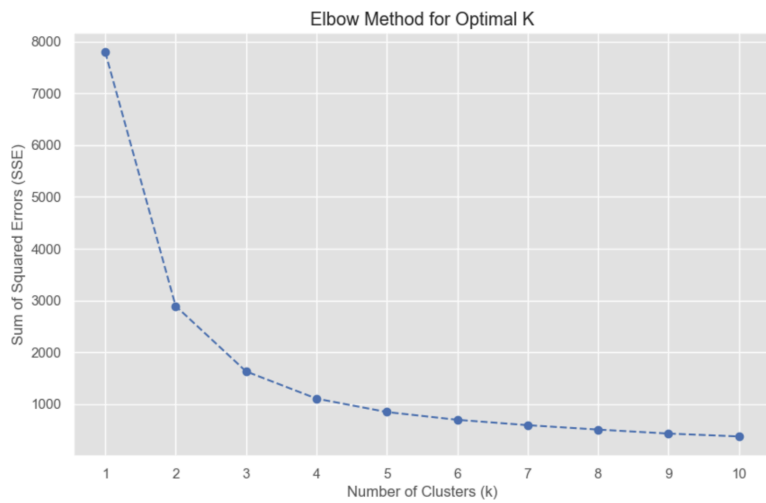
The methodology for this project followed a structured, multi-stage approach designed to transform raw transaction logs into high-level consumer intelligence. The process began with a rigorous data consolidation phase where 38,765 individual records were cleaned and grouped by both Member ID and Date, effectively merging disparate item purchases into 14,963 unique "baskets" or unified transactions.

	Member_number	Date	itemDescription
0	1808	2015-07-21	tropical fruit
1	2552	2015-01-05	whole milk
2	2300	2015-09-19	pip fruit
3	1187	2015-12-12	other vegetables
4	3037	2015-02-01	whole milk

Once the transactional structure was established, the project moved into association rule mining using the Apriori Algorithm. By applying a minimum support threshold of 0.001, the algorithm identified recurring combinations of products, which were then evaluated using statistical metrics like Support, Confidence, and Lift to determine the strength and predictability of specific product pairings.

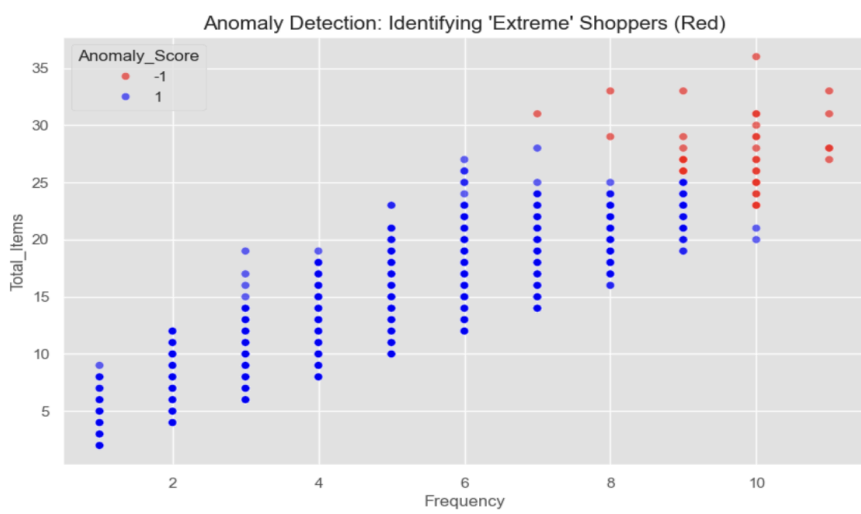


To move beyond aggregate patterns and understand individual consumer behavior, unsupervised machine learning techniques were applied to the member-level data. This involved implementing K-Means Clustering to segment the shopper base according to their visit frequency and total item volume. The optimal number of clusters was determined to be three ($k=3$) after a dual-validation process using the Elbow Method to find the point of diminishing returns in variance reduction and Silhouette Scores to measure the cohesion and separation of the resulting groups. As shown bellow.



The Elbow Method plot shows a distinct bend at $k = 3$. Beyond this point, the reduction in SSE becomes marginal, confirming that three clusters provide the most efficient balance between model complexity and data representation

Finally, to ensure the statistical integrity of these segments, an Isolation Forest algorithm was deployed for anomaly detection. This step successfully isolated "extreme" shoppers—outliers representing the top 1% of transaction volume—thereby protecting the final K-Means model from being skewed by non-human behaviors or wholesale accounts and ensuring a robust representation of the general customer population.




```
Detected 39 anomalous transactions.
```

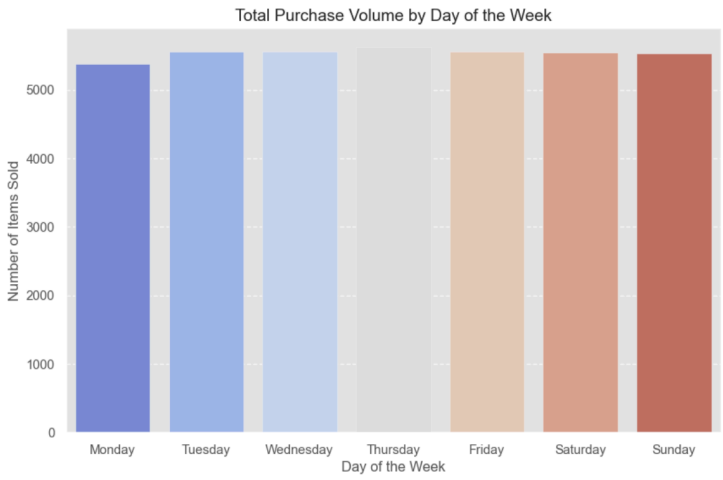
Using Isolation Forest, we identified {len(anomalies)} anomalous shoppers (red). These individuals represent the top 1% of transaction volume/frequency. Identifying these outliers ensures our K-Means clusters for Occasional, Regular, and VIP shoppers remain robust and aren't skewed by extreme edge cases.

3. Results

The exploratory data analysis phase provided a clear baseline for understanding the retail environment, revealing that the average basket size is remarkably small at only

2.59 items per transaction. This metric, coupled with the identification of top-performing "anchor" products. The five most frequently purchased items—whole milk, other vegetables, rolls and buns, soda, and yogurt—represent daily staples rather than luxury or specialty goods

Furthermore, temporal analysis showed that shopping volume remains remarkably consistent across all seven days of the week, lacking the dramatic spikes typically associated with "big weekend" grocery trips. These findings suggest the retailer functions as a daily necessity hub where customers perform frequent "top-off" visits for immediate household needs.



Building upon these foundational insights, the association rule mining phase utilized the Apriori algorithm to generate 240 distinct rules that define how products move together. The most significant relationship discovered was between yogurt and whole milk, which yielded a prominent Lift score of 2.18. This statistical measure indicates that a customer purchasing yogurt is more than twice as likely to include whole milk in their basket

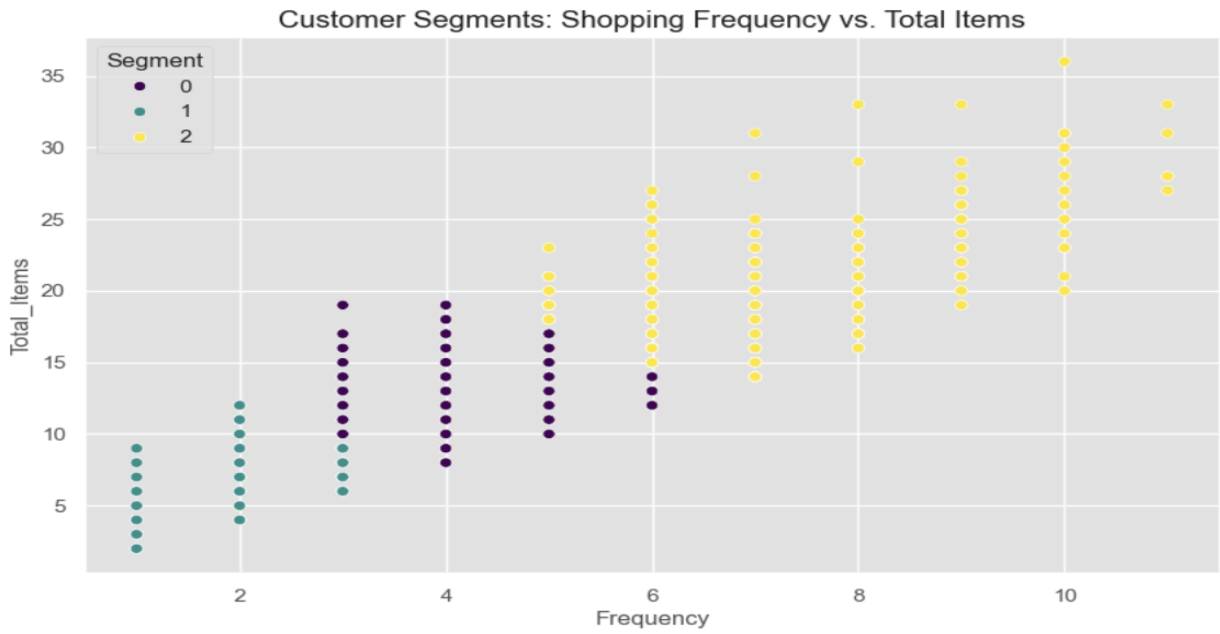
compared to the average shopper, representing a highly predictable and routine consumer behavior. Other meaningful pairings, such as specialty chocolate with citrus fruit and tropical fruit with flour, further illustrate specific niche purchasing patterns that exist alongside the primary dairy staples.

```
--- Final Mining Results: Top 10 Rules ---
```

Out[349…

	antecedents	consequents	support	confidence	lift
237	(sausage)	(yogurt, whole milk)	0.001470	0.024363	2.182917
236	(yogurt, whole milk)	(sausage)	0.001470	0.131737	2.182917
235	(sausage, whole milk)	(yogurt)	0.001470	0.164179	1.911760
238	(yogurt)	(sausage, whole milk)	0.001470	0.017121	1.911760
87	(specialty chocolate)	(citrus fruit)	0.001403	0.087866	1.653762
86	(citrus fruit)	(specialty chocolate)	0.001403	0.026415	1.653762
234	(sausage, yogurt)	(whole milk)	0.001470	0.255814	1.619866
239	(whole milk)	(sausage, yogurt)	0.001470	0.009310	1.619866
123	(tropical fruit)	(flour)	0.001069	0.015779	1.617141
122	(flour)	(tropical fruit)	0.001069	0.109589	1.617141

The final layer of analysis involved segmenting the customer base into three distinct behavioral profiles through K-Means clustering, allowing for a more granular view of loyalty and engagement. Segment 1, classified as "Occasional Shoppers," constitutes the group with the lowest engagement, averaging only 2.2 visits and a total of 5.4 items over the analyzed period. Segment 0, or "Regular Shoppers," maintains a moderate presence with an average of 4.4 visits and 11.5 total items. The most valuable group, Segment 2 or the "VIP Shoppers," demonstrates significantly higher loyalty, averaging 6.9 visits and purchasing 18.8 items on average. By identifying these segments, the retailer can transition from a one-size-fits-all approach to targeted strategies that address the specific needs and values of each shopper category.




```
Cluster Profiles (Mean values):
         Frequency  Total_Items
Segment
0         4.454201    11.487682
1         2.192666     5.423749
2         6.943049    18.864322
```

4. Interpretation

When viewed through the lens of retail management and consumer psychology, these results categorize the establishment as a "daily necessity hub." The high frequency of visits coupled with a consistently low basket size of 2.59 items reveals that the consumer base treats the store as a pantry extension rather than a primary fulfillment center. This behavior indicates a high level of trust and physical accessibility, where customers feel comfortable visiting for immediate, perishable needs like milk or vegetables rather than engaging in the long-term stockpiling typical of big-box retailers. The stability of transaction volume across all seven days of the week reinforces this "routine-based" shopping habit, suggesting that the store's revenue is not dependent on weekend surges but on the reliable, recurring needs of the local community.

The association rules provide a deeper look into the cognitive links between products, specifically the high Lift score of 2.18 between yogurt and whole milk. In a domain context, this suggests that dairy serves as the store’s primary "anchor" category, acting as the magnet that pulls customers through the doors. Because these items are predictable and routine, they provide a consistent baseline for inventory planning. However, the discovery of other pairings like specialty chocolates with citrus fruits or tropical fruits with flour suggests that while customers come in for staples, they are susceptible to cross-category impulse buys if items are positioned correctly. This

highlights an opportunity for "suggestive merchandising," where the store can leverage these hidden links to slowly increase the average basket size over time.

The most critical insight for long-term financial health is the identification of the "VIP Shopper" segment through K-Means clustering. This segment, though smaller in headcount than the "Occasional" group, drives a disproportionate percentage of total sales volume and visit frequency. In the retail industry, the cost of acquiring a new customer is significantly higher than retaining an existing one, making this group the store's most valuable asset. The interpretation of these clusters suggests that a "one-size-fits-all" marketing approach would be inefficient. Instead, the retailer should prioritize retention strategies for the VIPs, such as personalized loyalty rewards on anchor staples, while focusing on "conversion" strategies for the Regular and Occasional segments to migrate them toward higher frequency tiers.

5. Critical Assessment

While the analytical models employed in this project provided significant clarity regarding transaction patterns, a rigorous evaluation reveals several inherent limitations that must be considered before a full-scale implementation of the findings. One primary concern involves the parameters of the association rule mining; the decision to use a low minimum support threshold of 0.001 was necessary to uncover a broad range of relationships, but it introduces the risk of overestimating the importance of infrequent pairings. While certain itemsets exhibited high Lift scores—indicating a strong mathematical correlation—their low raw frequency means they may not occur often enough in the total transaction pool to justify expensive, store-wide physical restructuring. A strategic change based on such sparse data could lead to "over-optimization" for a tiny fraction of the customer base while potentially inconveniencing the majority of shoppers who follow more general patterns.

From a technical validity standpoint, the integration of the Isolation Forest algorithm was an essential safeguard to ensure the reliability of the customer segments. By identifying and flagging 39 "extreme" shoppers who represented the top 1% of transaction volume and frequency, the model successfully filtered out noise that could have skewed the cluster averages for Occasional, Regular, and VIP shoppers. This step was vital for preventing non-human accounts—such as automated bulk ordering or rare wholesale-style purchasing—from dictating a marketing strategy intended for local residents. Without this anomaly detection, the K-Means clusters would have likely reflected "average" behaviors that did not actually exist for any real subset of the population, thereby invalidating the resulting strategic recommendations.

Finally, the project faces a significant interpretative gap due to the total absence of demographic and psychographic data within the dataset. While the K-Means model

accurately identifies *how* different segments shop, it cannot explain *why* these behaviors occur. We can observe that Segment 2 visits the store nearly seven times more often than Segment 1, but without knowing the geographic proximity of these users to the store, their household size, or their income levels, we risk making over-generalized or biased assumptions about consumer intent. Ethically, this limitation serves as a reminder that data-driven personas are mathematical abstractions, not complete human profiles; relying solely on transaction volume for decision-making could inadvertently marginalize the needs of shoppers whose lower engagement is driven by socioeconomic factors rather than a lack of brand loyalty.

6. Conclusion

This comprehensive analysis has successfully mapped the consumer landscape of the retailer, transforming over 38,000 raw transaction records into a clear, strategic framework for business growth. The study identifies a convenience-driven operational model anchored by essential dairy and produce staples, where the store functions as a vital "pantry extension" for its local community. By leveraging exploratory data analysis, the project confirmed that while the average basket size remains small, the store maintains a remarkably steady and reliable growth trajectory that reached its peak in late 2015. This stability is primarily driven by a highly loyal "VIP" segment, whose frequent visits and higher volume signify a deep-rooted habituation with the store’s offerings.

The integration of advanced machine learning techniques, such as K-Means clustering and Isolation Forest anomaly detection, has provided the retailer with a robust roadmap for personalized engagement. The discovery of high-lift associations—most notably the strong link between yogurt and whole milk—offers a concrete starting point for immediate operational improvements. By recognizing that the shopper base is not a monolith but a collection of distinct behavioral profiles, the retailer is now equipped to move away from generic marketing and toward high-impact, segment-specific interventions that prioritize the retention of top-tier customers while seeking to deepen the engagement of occasional visitors.

Looking toward the future, the foundation laid by this project opens several avenues for further optimization. Future work should focus on integrating external datasets, weather patterns, or community events, to determine if external stressors or incentives can be leveraged to convert steady daily volume into larger, more profitable basket sizes. Additionally, the immediate implementation of targeted "bundle deals" or strategic aisle placements for identified high-lift item pairs is recommended as a low-cost, high-reward strategy. By continuing to apply data-driven insights to the store's physical and digital

presence, the retailer can refine its convenience-first identity into a more sophisticated, loyalty-centered business model.